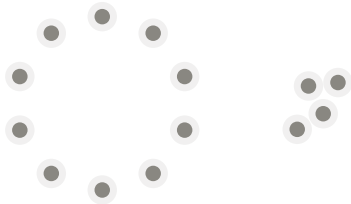


Persistent homology records shape across scale

As ε grows, components merge (H_0), loops appear (H_1), and short-lived features fade.

Small radius ε

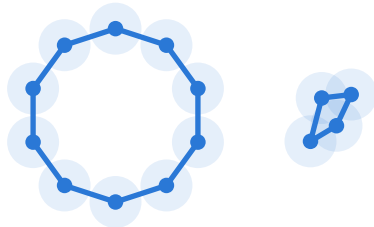
many separate components



H_0 : many components

Intermediate ε

nearby points connect

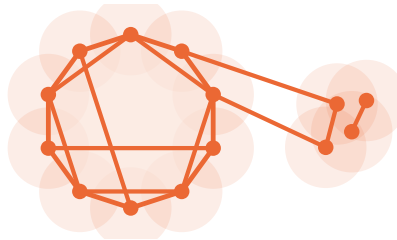


H_0 : 2 components

H_1 : 1 loop

Large radius ε

connections fill the loop



H_0 : 1 component

H_1 : loop filled

Persistence bars: long-lived features are the durable shape signal

H_0
 H_0
 H_1

Conceptual point-cloud illustration; this poster computes cubical persistence on packet images.

small ε

large ε